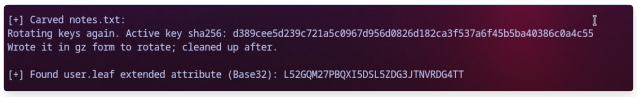
Fallen Leaf

Digital Forensics



I extracted `leafdisk.img` from `leafdisk.img.xz`. Rather than mounting the filesystem directly, I explored the disk structures using `debugfs` to preserve inode metadata.

Listing deleted inodes revealed a deleted file `harvest.enc` and a text file `notes.txt`. Carving `harvest.enc` recovered Part 1 of the flag: `d3l3t3d_k3ys_n3v3r_d13`.

Checking extended filesystem attributes on `notes.txt` revealed a hidden attribute `user.leaf` containing Base32 encoded data `L52GQM27PBQXI5DSL5ZDG3JTNVRDG4TT`. Decoding it yielded Part 2:`_th3_xattr_r3m3mb3rs`.

Finally, I tested the recovered flag against the executable / service:

`python3 Fallen_Leaf.py`

Part 1 (deleted inode) + Part 2 (user.leaf xattr) assembled.

Flag Captured

`SecLeaf{d3l3t3d_k3ys_n3v3r_d13_th3_xattr_r3m3mb3rs}`

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